

# Doc 11: Gradients and the Chain Rule

Module 2: Calculus | Part 1 of 3: Learn (theory). We build on Doc 10, one small step at a time.

## 1. The gradient: all the partial derivatives in one vector

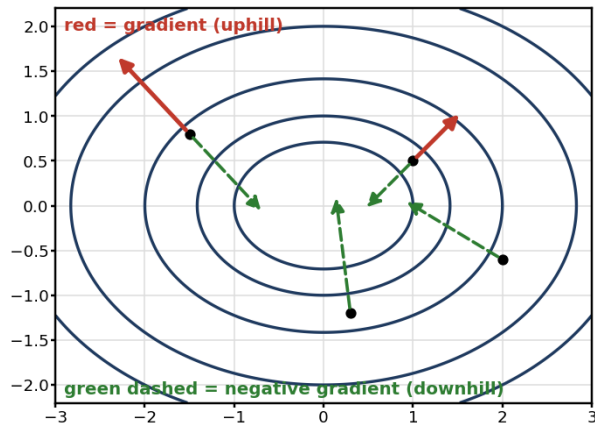
In Doc 10 you found  $\partial f/\partial x$  and  $\partial f/\partial y$  separately. The **gradient** stacks them into a single vector:  $\nabla f = (\partial f/\partial x, \partial f/\partial y)$ . ( $\nabla$  is read "nabla" or "del".) With more inputs, the vector just gets longer.

Example:  $f(x, y) = x^2 + 2y^2$  has  $\nabla f = (2x, 4y)$ . At the point  $(1, 0.5)$  it is  $(2, 2)$ .

### What the gradient tells you

- $\nabla f$  points **uphill**, in the direction where  $f$  increases fastest.
- Its **length** is how steep that climb is.
- It is **perpendicular to the contour lines** (the lines where  $f$  stays the same).
- $-\nabla f$  points steepest **downhill**: the basis of gradient descent.
- At a minimum, maximum, or flat spot,  $\nabla f = \mathbf{0}$ .

**Slope in any direction  $\mathbf{u}$**  (a unit vector):  $\nabla f \cdot \mathbf{u}$ , a dot product from Doc 4. It is biggest when  $\mathbf{u}$  lines up with  $\nabla f$ .



## 2. The chain rule: derivatives of functions inside functions

Take  $y = (3x + 1)^2$ . It is two steps: first  $\mathbf{u} = 3x + 1$  (inner), then  $\mathbf{y} = \mathbf{u}^2$  (outer). Each step has its own slope, and **slopes multiply**:

$$dy/dx = (dy/du) \times (du/dx) \text{ outer's slope} \times \text{inner's slope}$$

Here  $dy/du = 2u$  and  $du/dx = 3$ , so  $dy/dx = 2u \times 3 = 6(3x + 1)$ . Why they multiply: if  $u$  changes 3 times as fast as  $x$ , and  $y$  changes  $2u$  times as fast as  $u$ , then  $y$  changes  $3 \times 2u$  times as fast as  $x$ .

# Doc 11: Gradients and the Chain Rule

Part 1 of 3: Learn (theory), continued

## 3. A routine for the chain rule (use it every time)

- **Step 1.** Name the inner function  $u$  (the part inside brackets, an exponent, or a log).
- **Step 2.** Differentiate the **outer** function with respect to  $u$ .
- **Step 3.** Differentiate the **inner** function with respect to  $x$ .
- **Step 4.** Multiply the two, then put  $u$  back in terms of  $x$ .

|                    |                                              |                             |
|--------------------|----------------------------------------------|-----------------------------|
| Bracket to a power | $d/dx [g(x)]^n = n [g(x)]^{n-1} \cdot g'(x)$ | $(3x+1)^2: 2(3x+1) \cdot 3$ |
| Exponential        | $d/dx e^{g(x)} = e^{g(x)} \cdot g'(x)$       | $e^{2x}: e^{2x} \cdot 2$    |
| Logarithm          | $d/dx \ln g(x) = g'(x) / g(x)$               | $\ln(x^2+1): 2x / (x^2+1)$  |

## 4. Three derivatives every neural network uses

| Activation       | Formula                           | Derivative                        |
|------------------|-----------------------------------|-----------------------------------|
| ReLU             | $\max(0, x)$                      | 0 for $x < 0$ , and 1 for $x > 0$ |
| Sigmoid $\sigma$ | $1 / (1 + e^{-x})$                | $\sigma(x) (1 - \sigma(x))$       |
| Tanh             | $(e^x - e^{-x}) / (e^x + e^{-x})$ | $1 - \tanh(x)^2$                  |

**Sigmoid derived with the chain rule:**  $\sigma = (1 + e^{-x})^{-1}$ . Inner  $u = 1 + e^{-x}$ ,  $du/dx = -e^{-x}$ . Outer  $u^{-1}$  has slope  $-u^{-2}$ . So  $\sigma' = -u^{-2} \times (-e^{-x}) = e^{-x} / (1 + e^{-x})^2 = \sigma (1 - \sigma)$ . The slope peaks at 0.25 when  $x = 0$  and fades to nearly 0 for large  $|x|$ : the **vanishing gradient** problem.

## 5. The gradient of the squared error (the most important formula in this doc)

A line predicts  $\hat{y} = w x + b$ . The error is  $e = \hat{y} - y$ , and the loss is  $L = e^2$ . By the chain rule (outer  $e^2$  has slope  $2e$ ; inner  $e$  has slopes  $x$  for  $w$  and 1 for  $b$ ):

$$\partial L / \partial w = 2 e \times \partial L / \partial b = 2 e$$

**Gradient descent step:** the new weights are  $(w, b)$  minus  $\eta$  times  $\nabla L = (2ex, 2e)$ . If the prediction is too low ( $e$  negative) the weights rise; too high ( $e$  positive) they fall.

## 6. Where machine learning uses these

- **Training** means computing  $\nabla L$  for every weight, then stepping against it.
- **Backpropagation** (Doc 24) is the chain rule applied layer by layer: multiply local slopes from the loss back to each weight (Example 6).
- **Vanishing and exploding gradients:** a long product of slopes below 1 shrinks toward 0; above 1 it blows up.
- **Gradient checking:** compare your gradient with  $(L(w + h) - L(w)) / h$ .

# Doc 11: Gradients and the Chain Rule

Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

## Example 1: A gradient, its length, and the slope in different directions

$f(x, y) = x^2 + 3xy$ .  $\partial f / \partial x = 2x + 3y$  (treat  $y$  as constant) and  $\partial f / \partial y = 3x$ . So  $\nabla f = (2x + 3y, 3x)$ .

| Quantity                            | Work                                                         | Value                 |
|-------------------------------------|--------------------------------------------------------------|-----------------------|
| $\nabla f$ at (1, 2)                | $(2 \cdot 1 + 3 \cdot 2, 3 \cdot 1)$                         | <b>(8, 3)</b>         |
| Length $ \nabla f $                 | $\text{sqrt}(8^2 + 3^2) = \text{sqrt}(73)$                   | <b>8.544</b>          |
| Slope going right, $u = (1, 0)$     | $\nabla f \cdot u = 8 \times 1 + 3 \times 0$                 | <b>8</b>              |
| Slope in direction $u = (0.6, 0.8)$ | $\nabla f \cdot u = 8 \times 0.6 + 3 \times 0.8 = 4.8 + 2.4$ | <b>7.2</b>            |
| Steepest uphill direction           | $\nabla f /  \nabla f  = (8, 3) / 8.544$                     | <b>(0.936, 0.351)</b> |

Every direction gives a slope no bigger than the gradient's length 8.544. Only the direction of  $\nabla f$  itself reaches it.

## Example 2: The chain rule with a table of pieces

| Function       | Inner $u$ | $du/dx$                | $dy/du$                   | $dy/dx = dy/du \times du/dx$                    |
|----------------|-----------|------------------------|---------------------------|-------------------------------------------------|
| $(3x + 1)^2$   | $3x + 1$  | <b>3</b>               | <b><math>2u</math></b>    | <b><math>2(3x+1) \cdot 3 = 6(3x + 1)</math></b> |
| $e^{2x}$       | $2x$      | <b>2</b>               | <b><math>e^u</math></b>   | <b><math>2e^{2x}</math></b>                     |
| $\ln(x^2 + 1)$ | $x^2 + 1$ | <b><math>2x</math></b> | <b><math>1 / u</math></b> | <b><math>2x / (x^2 + 1)</math></b>              |
| $(x^2 + 1)^3$  | $x^2 + 1$ | <b><math>2x</math></b> | <b><math>3u^2</math></b>  | <b><math>6x (x^2 + 1)^2</math></b>              |

**Check with numbers:**  $y = (3x + 1)^2$  at  $x = 1$  has slope  $6 \times 4 = 24$ . Now  $f(1.01) = 4.03^2 = 16.2409$ ,  $f(1) = 16$ , and  $(16.2409 - 16) / 0.01 = 24.09 \approx 24$  ✓

## Example 3: The sigmoid's slope at a few points, using $\sigma' = \sigma(1 - \sigma)$

| $x$       | $\sigma(x)$  | $\sigma'(x) = \sigma(1 - \sigma)$              |
|-----------|--------------|------------------------------------------------|
| <b>-2</b> | <b>0.119</b> | <b><math>0.119 \times 0.881 = 0.105</math></b> |
| <b>0</b>  | <b>0.5</b>   | <b><math>0.5 \times 0.5 = 0.25</math></b>      |
| <b>1</b>  | <b>0.731</b> | <b><math>0.731 \times 0.269 = 0.197</math></b> |
| <b>2</b>  | <b>0.881</b> | <b><math>0.881 \times 0.119 = 0.105</math></b> |

The slope is biggest (0.25) at  $x = 0$  and is only 0.105 at  $x = \pm 2$ . Stack many sigmoid layers and multiply slopes like these: the product shrinks fast.

# Doc 11: Gradients and the Chain Rule

Part 2 of 3: Worked examples, continued

## Example 4: The picture of a gradient step (the gradient of $x^2 + 2y^2$ )

On the picture on page 1,  $\nabla f = (2x, 4y)$ . At  $(2, -0.6)$  it is  $(4, -2.4)$ : it points away from the center, across the contours. One descent step with  $\eta = 0.1$  moves to  $(2, -0.6) - 0.1 \times (4, -2.4) = (1.6, -0.36)$ , toward the center. Check:  $f$  before  $= 4 + 0.72 = 4.72$ ;  $f$  after  $= 2.56 + 0.259 = 2.82$ . It went down.

## Example 5: Gradient of the squared error, then one descent step

One data point:  $x = 2, y = 5$ . Current line:  $w = 1, b = 0$ . Step size  $\eta = 0.05$ .

| Quantity                                          | Before the step       | After the step             |
|---------------------------------------------------|-----------------------|----------------------------|
| Prediction $\hat{y} = w x + b$                    | $1 \times 2 + 0 = 2$  | $1.6 \times 2 + 0.3 = 3.5$ |
| Error $e = \hat{y} - y$                           | $2 - 5 = -3$          | $3.5 - 5 = -1.5$           |
| Loss $L = e^2$                                    | 9                     | 2.25                       |
| $\partial L / \partial w = 2 e x$                 | $2(-3)(2) = -12$      |                            |
| $\partial L / \partial b = 2 e$                   | $2(-3) = -6$          |                            |
| New $w = w - \eta \times \partial L / \partial w$ | $1 - 0.05(-12) = 1.6$ |                            |
| New $b = b - \eta \times \partial L / \partial b$ | $0 - 0.05(-6) = 0.3$  |                            |

Both gradients are negative (the prediction is too low), so both  $w$  and  $b$  rise. The loss fell from 9 to 2.25 in one step.

## Example 6: Backpropagation by hand (forward pass, then backward pass)

A one-neuron model:  $z = w x + b$ ,  $a = \text{ReLU}(z)$ , loss  $L = (a - t)^2$ . Data:  $x = 2$ , target  $t = 5$ . Weights:  $w = 3, b = 1$ .

| Pass     | Quantity                                                                           | Work             | Value |
|----------|------------------------------------------------------------------------------------|------------------|-------|
| Forward  | $z = w x + b$                                                                      | $3 \times 2 + 1$ | 7     |
| Forward  | $a = \text{ReLU}(z)$                                                               | $\max(0, 7)$     | 7     |
| Forward  | $L = (a - t)^2$                                                                    | $(7 - 5)^2$      | 4     |
| Backward | $\partial L / \partial a = 2(a - t)$                                               | $2(7 - 5)$       | 4     |
| Backward | $\partial a / \partial z$ (ReLU slope)                                             | $z = 7 > 0$      | 1     |
| Backward | $\partial L / \partial z = \partial L / \partial a \times \partial a / \partial z$ | $4 \times 1$     | 4     |
| Backward | $\partial L / \partial w = \partial L / \partial z \times x$                       | $4 \times 2$     | 8     |
| Backward | $\partial L / \partial b = \partial L / \partial z \times 1$                       | $4 \times 1$     | 4     |

Update with  $\eta = 0.1$ :  $w = 3 - 0.8 = 2.2$ ,  $b = 1 - 0.4 = 0.6$ . New  $z = 2.2 \times 2 + 0.6 = 5$ ,  $a = 5$ ,  $L = 0$ . The gradient led straight to a perfect fit. If  $z$  had been negative,  $\partial a / \partial z = 0$  would zero every gradient: a "dead" ReLU.

# Doc 11: Gradients and the Chain Rule

Part 2 of 3: Worked examples, continued

## Example 7: The mean gradient over a batch

Two data points:  $(x = 1, y = 3)$  and  $(x = 2, y = 5)$ . Current  $w = 0$ ,  $b = 0$ . The batch loss is the **mean** of the squared errors, and so is its gradient.

| Point | x | y | y-hat = w x + b | e = y-hat - y | e <sup>2</sup> | 2 e x | 2 e |
|-------|---|---|-----------------|---------------|----------------|-------|-----|
| 1     | 1 | 3 | 0               | -3            | 9              | -6    | -6  |
| 2     | 2 | 5 | 0               | -5            | 25             | -20   | -10 |
| Mean  |   |   |                 |               | 17             | -13   | -8  |

Batch loss = 17.  $\nabla L = (\partial L / \partial w, \partial L / \partial b) = (-13, -8)$ . With  $\eta = 0.05$ :  $w = 0 + 0.65 = 0.65$  and  $b = 0 + 0.4 = 0.4$ . New predictions: 1.05 and 1.7, both closer to 3 and 5.

## Common mistakes

| Mistake                                                  | Correct                                   |
|----------------------------------------------------------|-------------------------------------------|
| $d/dx (3x + 1)^2 = 2(3x + 1)$ (forgot the inner slope 3) | $6(3x + 1)$                               |
| $d/dx e^{2x} = e^{2x}$ (forgot the inner slope 2)        | $2e^{2x}$                                 |
| $d/dx \ln(x^2 + 1) = 1 / (x^2 + 1)$ (forgot the 2x)      | $2x / (x^2 + 1)$                          |
| The gradient is a single number                          | It is a vector with one entry per input   |
| Gradient descent: $w = w + \eta \nabla L$                | $w = w - \eta \nabla L$ (minus: downhill) |

## Quick reference card

|                        |                                                                            |                                                               |
|------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------|
| Gradient               | $\nabla f = (\partial f / \partial x_1, \partial f / \partial x_2, \dots)$ | uphill; $-\nabla f$ is downhill; $\nabla f = 0$ at flat spots |
| Slope in direction $u$ | $\nabla f \cdot u$                                                         | largest = $ \nabla f $ when $u$ lines up with $\nabla f$      |
| Chain rule             | $dy/dx = dy/du \times du/dx$                                               | outer slope $\times$ inner slope                              |
| Sigmoid and ReLU       | $\sigma' = \sigma(1 - \sigma)$ ; $\text{ReLU}' = 0$ or $1$                 | $\tanh' = 1 - \tanh^2$                                        |
| Squared error          | $\partial L / \partial w = 2 e x$ , $\partial L / \partial b = 2 e$        | $e = y\text{-hat} - y$                                        |
| Descent step           | $w$ becomes $w - \eta \partial L / \partial w$                             | minus sign: downhill                                          |

# Doc 11: Gradients and the Chain Rule

Part 3 of 3: Practice. Yellow cells are yours. Name: \_\_\_\_\_ Date: \_\_\_\_\_

## A. Gradients

Find each partial derivative, then write the gradient vector at the given point.

| Function and point                  | $\partial f / \partial x$ | $\partial f / \partial y$ | $\nabla f$ at the point |
|-------------------------------------|---------------------------|---------------------------|-------------------------|
| 1. $f = x^2 + y^2$ , point (3, -1)  |                           |                           |                         |
| 2. $f = 3x + 2y - 5$ , point (4, 7) |                           |                           |                         |
| 3. $f = x^2y$ , point (2, 3)        |                           |                           |                         |
| 4. $f = xy + y^2$ , point (1, 2)    |                           |                           |                         |
| 5. $f = e^x + y^3$ , point (0, 1)   |                           |                           |                         |

## B. Slope in a direction: $f = x^2 + y^2$ at (3, 4)

First  $\nabla f = (2x, 2y) = ( \quad , \quad )$ . Then the slope in direction  $u$  is  $\nabla f \cdot u$ .

| Direction $u$ | $\nabla f \cdot u$ | What it means |
|---------------|--------------------|---------------|
| (1, 0)        |                    |               |
| (0, 1)        |                    |               |
| (0.6, 0.8)    |                    |               |
| (-0.6, -0.8)  |                    |               |
| (0.8, -0.6)   |                    |               |

## C. The chain rule with a table of pieces

Fill the inner  $u$ , then  $du/dx$ , then  $dy/du$  (in terms of  $u$ ), then multiply.

| Function              | Inner $u$ | $du/dx$ | $dy/du$ | $dy/dx$ |
|-----------------------|-----------|---------|---------|---------|
| $(2x + 1)^3$          |           |         |         |         |
| $e^{3x}$              |           |         |         |         |
| $\ln(2x)$             |           |         |         |         |
| $(x^2 - 1)^4$         |           |         |         |         |
| $\text{sqrt}(4x + 1)$ |           |         |         |         |
| $1 / (x + 2)$         |           |         |         |         |

# Doc 11: Gradients and the Chain Rule

Practice, continued

## D. Check a chain-rule answer with numbers

$y = (3x + 1)^2$  at  $x = 2$ . Rule:  $dy/dx = 6(3x + 1) = \underline{\hspace{2cm}}$ . Numeric check with  $h = 0.01$ :

| $f(2)$ | $f(2.01)$ | $f(2.01) - f(2)$ | Slope = difference / 0.01 |
|--------|-----------|------------------|---------------------------|
| 49     |           |                  |                           |

## E. The sigmoid's slope: $\sigma' = \sigma(1 - \sigma)$

| $x$ | $\sigma(x)$ (given) | $1 - \sigma(x)$ | $\sigma'(x)$ |
|-----|---------------------|-----------------|--------------|
| -1  | 0.269               |                 |              |
| 0   | 0.5                 |                 |              |
| 1   | 0.731               |                 |              |
| 3   | 0.953               |                 |              |

The slope is largest at  $x = \underline{\hspace{2cm}}$ . For large  $|x|$  the slope is close to  $\underline{\hspace{2cm}}$ : this is the  $\underline{\hspace{2cm}}$  gradient problem.

## F. Gradient of the squared error, then a descent step

One data point:  $x = 3$ ,  $y = 7$ . Current line:  $w = 1$ ,  $b = 1$ . Step size  $\eta = 0.02$ .

| Quantity                                          | Before the step | After the step |
|---------------------------------------------------|-----------------|----------------|
| Prediction $\hat{y} = w x + b$                    |                 |                |
| Error $e = \hat{y} - y$                           |                 |                |
| Loss $L = e^2$                                    |                 |                |
| $\partial L / \partial w = 2 e x$                 |                 |                |
| $\partial L / \partial b = 2 e$                   |                 |                |
| New $w = w - \eta \times \partial L / \partial w$ |                 |                |
| New $b = b - \eta \times \partial L / \partial b$ |                 |                |

Did the loss go down?  $\underline{\hspace{2cm}}$

# Doc 11: Gradients and the Chain Rule

Practice, continued

## G. The mean gradient over a batch

Points  $(x = 1, y = 2)$  and  $(x = 3, y = 4)$ . Current  $w = 1, b = 0$ . Fill the table; the last row is the mean of the rows above.

| Point | x | y | y-hat = w x + b | e = y-hat - y | e <sup>2</sup> | 2 e x | 2 e |
|-------|---|---|-----------------|---------------|----------------|-------|-----|
| 1     | 1 | 2 |                 |               |                |       |     |
| 2     | 3 | 4 |                 |               |                |       |     |
| Mean  |   |   |                 |               |                |       |     |

Batch loss = \_\_\_\_\_.  $\nabla L = ( \text{_____} , \text{_____} )$ . With  $\eta = 0.1$ : new  $w = \text{_____}$ , new  $b = \text{_____}$ .

## H. Backpropagation by hand

Model:  $z = w x + b$ ,  $a = \text{ReLU}(z)$ ,  $L = (a - t)^2$ . Data:  $x = 3$ , target  $t = 2$ . Weights:  $w = 2, b = -1$ .

| Pass     | Quantity                                                                           | Your work | Value |
|----------|------------------------------------------------------------------------------------|-----------|-------|
| Forward  | $z = w x + b$                                                                      |           |       |
| Forward  | $a = \text{ReLU}(z)$                                                               |           |       |
| Forward  | $L = (a - t)^2$                                                                    |           |       |
| Backward | $\partial L / \partial a = 2(a - t)$                                               |           |       |
| Backward | $\partial a / \partial z$ (ReLU slope)                                             |           |       |
| Backward | $\partial L / \partial z = \partial L / \partial a \times \partial a / \partial z$ |           |       |
| Backward | $\partial L / \partial w = \partial L / \partial z \times x$                       |           |       |
| Backward | $\partial L / \partial b = \partial L / \partial z \times 1$                       |           |       |

Bonus: if  $z$  had been -2 instead, what would  $\partial a / \partial z$ ,  $\partial L / \partial w$  and  $\partial L / \partial b$  be? \_\_\_\_\_ What is this problem called? \_\_\_\_\_

# Doc 11: Gradients and the Chain Rule

Practice, continued: more repetitions

## I. Six more chain-rule derivatives

Name the inner function  $u$  first, then use the routine: outer slope  $\times$  inner slope.

| Function          | Inner $u$ | $dy/dx$ |
|-------------------|-----------|---------|
| $(5x - 2)^2$      |           |         |
| $e^{-x}$          |           |         |
| $\ln(1 + x^2)$    |           |         |
| $(x^2 + 3x)^2$    |           |         |
| $e^{x^2}$         |           |         |
| $\text{sqrt}(2x)$ |           |         |

## J. True or False?

| Statement                                                          | True / False |
|--------------------------------------------------------------------|--------------|
| 1. The gradient points in the direction of steepest increase.      |              |
| 2. At a minimum of $f$ the gradient is the zero vector.            |              |
| 3. The gradient is parallel to the contour lines.                  |              |
| 4. By the chain rule, $d/dx (e^{2x}) = e^{2x}$ .                   |              |
| 5. The sigmoid's slope is largest at $x = 0$ .                     |              |
| 6. Gradient descent adds $\eta$ times the gradient to the weights. |              |

## K. Self-check

| I can...                                    | Revisit if unsure         | Confident? |
|---------------------------------------------|---------------------------|------------|
| write a gradient from partial derivatives   | Example 1, Exercise A     |            |
| find a slope in a direction (dot product)   | Example 1, Exercise B     |            |
| apply the chain rule with a table           | Example 2, Exercises C, I |            |
| use the sigmoid and ReLU derivatives        | Example 3, Exercise E     |            |
| compute a squared-error gradient and a step | Example 5, Exercise F     |            |
| do a forward and backward pass by hand      | Example 6, Exercise H     |            |

# Doc 11: Gradients and the Chain Rule

Answer key (page 1 of 3): check your work only after finishing

## A. Gradients

| Function and point                  | $\partial f / \partial x$ | $\partial f / \partial y$ | $\nabla f$ at the point |
|-------------------------------------|---------------------------|---------------------------|-------------------------|
| 1. $f = x^2 + y^2$ , point (3, -1)  | 2x                        | 2y                        | (6, -2)                 |
| 2. $f = 3x + 2y - 5$ , point (4, 7) | 3                         | 2                         | (3, 2)                  |
| 3. $f = x^2y$ , point (2, 3)        | 2xy                       | $x^2$                     | (12, 4)                 |
| 4. $f = xy + y^2$ , point (1, 2)    | y                         | $x + 2y$                  | (2, 5)                  |
| 5. $f = e^x + y^3$ , point (0, 1)   | $e^x$                     | $3y^2$                    | (1, 3)                  |

## B. Slope in a direction

$\nabla f = (6, 8)$ , length 10.

| Direction u  | $\nabla f \cdot u$ | Meaning                                            |
|--------------|--------------------|----------------------------------------------------|
| (1, 0)       | 6                  | slope going right                                  |
| (0, 1)       | 8                  | slope going up                                     |
| (0.6, 0.8)   | 10                 | steepest uphill: equals the length of the gradient |
| (-0.6, -0.8) | -10                | steepest downhill                                  |
| (0.8, -0.6)  | 0                  | along the contour: no change                       |

(1, 0):  $6 \times 1 = 6$ . (0, 1): 8. (0.6, 0.8):  $3.6 + 6.4 = 10$ . (-0.6, -0.8): -10. (0.8, -0.6):  $4.8 - 4.8 = 0$ .

## C. The chain rule

| Function              | Inner u   | du/dx | dy/du                  | dy/dx                     |
|-----------------------|-----------|-------|------------------------|---------------------------|
| $(2x + 1)^3$          | $2x + 1$  | 2     | $3u^2$                 | $6(2x + 1)^2$             |
| $e^{3x}$              | 3x        | 3     | $e^u$                  | $3e^{3x}$                 |
| $\ln(2x)$             | 2x        | 2     | $1/u$                  | $2 / (2x) = 1/x$          |
| $(x^2 - 1)^4$         | $x^2 - 1$ | 2x    | $4u^3$                 | $8x(x^2 - 1)^3$           |
| $\text{sqrt}(4x + 1)$ | $4x + 1$  | 4     | $\frac{1}{2} u^{-1/2}$ | $2 / \text{sqrt}(4x + 1)$ |
| $1 / (x + 2)$         | $x + 2$   | 1     | $-u^{-2}$              | $-1 / (x + 2)^2$          |

## D. Numeric check

$dy/dx = 6(3 \cdot 2 + 1) = 6 \times 7 = 42$ .  $f(2.01) = 7.03^2 = 49.4209$ , difference 0.4209, slope  $0.4209 / 0.01 = 42.09 \approx 42$   
✓

## E. Sigmoid slope

| x  | $\sigma(x)$ | $1 - \sigma(x)$ | $\sigma'(x)$ |
|----|-------------|-----------------|--------------|
| -1 | 0.269       | 0.731           | 0.197        |
| 0  | 0.5         | 0.5             | 0.25         |
| 1  | 0.731       | 0.269           | 0.197        |
| 3  | 0.953       | 0.047           | 0.045        |

Largest at  $x = 0$  (0.25). For large  $|x|$  the slope is close to 0: the vanishing gradient problem.

# Doc 11: Gradients and the Chain Rule

Answer key (page 2 of 3)

## F. Squared error and a descent step

| Quantity                                          | Before the step        | After the step               |
|---------------------------------------------------|------------------------|------------------------------|
| Prediction $\hat{y} = w x + b$                    | $1 \times 3 + 1 = 4$   | $1.36 \times 3 + 1.12 = 5.2$ |
| Error $e = \hat{y} - y$                           | $4 - 7 = -3$           | $5.2 - 7 = -1.8$             |
| Loss $L = e^2$                                    | 9                      | 3.24                         |
| $\partial L / \partial w = 2 e x$                 | $2(-3)(3) = -18$       | -                            |
| $\partial L / \partial b = 2 e$                   | $2(-3) = -6$           | -                            |
| New $w = w - \eta \times \partial L / \partial w$ | $1 - 0.02(-18) = 1.36$ | -                            |
| New $b = b - \eta \times \partial L / \partial b$ | $1 - 0.02(-6) = 1.12$  | -                            |

Yes: the loss went from 9 down to 3.24.

## G. The mean gradient over a batch

| Point | x | y | y-hat | e  | e <sup>2</sup> | 2 e x | 2 e |
|-------|---|---|-------|----|----------------|-------|-----|
| 1     | 1 | 2 | 1     | -1 | 1              | -2    | -2  |
| 2     | 3 | 4 | 3     | -1 | 1              | -6    | -2  |
| Mean  |   |   |       |    | 1              | -4    | -2  |

Batch loss 1.  $\nabla L = (-4, -2)$ . New  $w = 1 - 0.1(-4) = 1.4$  and new  $b = 0 - 0.1(-2) = 0.2$ . (y-hat =  $w x + b$  with  $w = 1$ ,  $b = 0$  gives 1 and 3.)

## H. Backpropagation by hand

| Pass     | Quantity                                                     | Work             | Value |
|----------|--------------------------------------------------------------|------------------|-------|
| Forward  | $z = w x + b$                                                | $2 \times 3 - 1$ | 5     |
| Forward  | $a = \text{ReLU}(z)$                                         | $\max(0, 5)$     | 5     |
| Forward  | $L = (a - t)^2$                                              | $(5 - 2)^2$      | 9     |
| Backward | $\partial L / \partial a = 2(a - t)$                         | $2(5 - 2)$       | 6     |
| Backward | $\partial a / \partial z$ (ReLU slope)                       | $z = 5 > 0$      | 1     |
| Backward | $\partial L / \partial z$                                    | $6 \times 1$     | 6     |
| Backward | $\partial L / \partial w = \partial L / \partial z \times x$ | $6 \times 3$     | 18    |
| Backward | $\partial L / \partial b = \partial L / \partial z \times 1$ | $6 \times 1$     | 6     |

Bonus: with  $z = -2$ , ReLU's slope is 0, so  $\partial a / \partial z = 0$  and both  $\partial L / \partial w$  and  $\partial L / \partial b$  are 0. This is the **dead ReLU** problem: no learning signal reaches  $w$  or  $b$ .

## I. Six more chain-rule derivatives

| Function          | Inner u    | dy/dx                                                    |
|-------------------|------------|----------------------------------------------------------|
| $(5x - 2)^2$      | $5x - 2$   | $2(5x - 2) \times 5 = 10(5x - 2)$                        |
| $e^{-x}$          | $-x$       | $e^{-x} \times (-1) = -e^{-x}$                           |
| $\ln(1 + x^2)$    | $1 + x^2$  | $2x / (1 + x^2)$                                         |
| $(x^2 + 3x)^2$    | $x^2 + 3x$ | $2(x^2 + 3x)(2x + 3)$                                    |
| $e^{x^2}$         | $x^2$      | $e^{x^2} \times 2x$                                      |
| $\text{sqrt}(2x)$ | $2x$       | $\frac{1}{2} (2x)^{-1/2} \times 2 = 1 / \text{sqrt}(2x)$ |

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Answer key (page 3 of 3)

## J. True or False?

1 True. 2 True. 3 **False** (the gradient is perpendicular to the contour lines). 4 **False** (it is  $2e^{2x}$ : the inner slope 2). 5 True. 6 **False** (subtract:  $w$  becomes  $w - \eta \nabla L$ ).

## K. Self-check

No wrong answers: any 'Not yet' points you to the example and exercise to redo. Practice again after a day or two.

**Key takeaways:**  $\nabla f$  is the vector of partial derivatives: it points uphill, its length is the steepness, and  $-\nabla f$  is the way down; the slope in direction  $u$  is  $\nabla f \cdot u$ ; the chain rule multiplies outer slope by inner slope;  $\sigma' = \sigma(1 - \sigma)$ ; for squared error  $\partial L / \partial w = 2 e^x$  and  $\partial L / \partial b = 2 e$ ; backpropagation multiplies local slopes from the loss backward.

**Next: Doc 12** - Jacobian and Hessian.